

Assessment #4 - Research

Student Name: Robin Kim

ISM I

Date: 9/8/26

Subject: Assessment #4 - Research

APA Citation:

Scarborough, D. M., Colón, P. E., Linderman, S. E., & Berkson, E. M. (2021). Comparison of kinematic sequences during curveball and fastball baseball pitches. *Frontiers in Sports and Active Living*, 3, Article 699251. <https://doi.org/10.3389/fspor.2021.699251>

Entry

Scarborough et al. (2021) examined kinematic sequencing (KS), the order in which the pelvis(1), trunk(2), arm(3), forearm(4), hand(5), and maximum angular velocity across 71 curveball pitches thrown by 14 pitchers ranging from high school to professional level. It then compared these patterns to fastball kinetic sequencings from the same group. The study examined 11 distinct curveball KS patterns and figured out that none of the 71 pitches was thrown in ideal "proximal-to-distal" sequence; instead, the two most common patterns showed altered distal upper extremity (DUE) timing depending on elbow and shoulder torque (Scarborough et al., 2021).

What I found most relevant to my ISM research is that this study tests hypotheses across competition levels rather than blindly assuming professional mechanics scale downward, which is exactly the kind of point that needs to be made in my project. My research question asks to what extent MLB-level sabermetric strategies apply to high school baseball, and Scarborough et al. (2021) essentially ask the same question but in terms of biomechanics: does the "ideal" KS taught to pitchers at every level actually hold up when measured?

The findings, that 0 out of 71 pitches achieved the target sequence regardless of the competition, tells me that the discrepancies between theoretical mechanics and live pitching is not merely a beginner's problem that gradually disappears. That directly supports the skepticism that raw theoretical models designed around professional performance cannot simply be dropped onto high school data without further examination of contextual constraints but calibrate them accordingly.

Evaluating the study's methodology, I noticed real limitations that temper how far I can extend its conclusions. The sample was only 14 pitches with 71 pitches (~5 pitches per pitcher) collected in a single lab session, which the authors themselves acknowledge cannot capture the temporal variable like fatigue, mound conditions, or game pressure in real games. Additionally, there is a potential detection bias that the 360Hz camera system may not distinguish genuinely simultaneous velocities from peaks too close to resolve, which the authors admit in their limitation section. This uncertainty matters because some of the data used in this is vulnerable to near-simultaneous timings between the forearm and hand.

One gap the article leaves unresolved is whether the altered kinematic sequences shown to increase joint torque in fastballs — 12353 and 12354 patterns — produce the same amount of

torque increases during curveballs. The authors note that this is unknown and leave it a direction for future research. That open question is where I see the most relevancy and connection to my potential original work. My predictive model on umpire strike-call tendencies focuses on pitch location outcomes, and this article's point of how timing errors are the hardest for coaches to capture on the spot, especially during the curveball's characteristic forearm supination (top-down), suggests a synthesis I had not previously considered: inconsistent KS at HS level could be a mechanical driver behind the command and location variability that is already being fed as input to my strike-call predictions. Rather than treating command as an original outcome-based statistic, I could use this article to suggest that command variability at HS level has a biomechanical origin. Ultimately, this literature helped me find a genuinely original angle for my research, and it reinforces why my project cannot just scale down MLB analytics, but has to be built with an additional layer for the movement variability specific to less experienced throwers.

Kami Link

<https://kami.app/wj8-e6E-7Qu-jHs>